InboxRL: Pure Reinforcement Learning (GRPO) for Calendar Event Extraction from Institute Emails

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Code: https://github.com/hhemantpatel/inboxrl

***Abstract*—We train a small language model (Qwen2.5-0.5B-Instruct with LoRA) to turn an institute email into a JSON list of calendar events *purely with reinforcement learning*: no supervised fine-tuning, and labels are used only inside a verifiable reward. Generation is a token-level MDP optimised with GRPO. The reward combines a format gate, a Hungarian match between predicted and gold events, and a calendar-arithmetic date verifier. On 300 template-disjoint synthetic test emails the best run reaches event F1 72.2% (95% CI 67.4–77.2), against 24.0% for the untrained 0.5B model and 67.1% for a 3× larger zero-shot model. The main finding is a failure mode: the ability to answer "no events" ([]) collapsed in three configurations.**

I. Problem

Deadlines and quizzes reach students as free-form emails. Extraction must resolve relative dates ("this Thursday" from the received date d_0), ignore superseded ones, handle several events and noisy text, return nothing when there is no event, and emit JSON that exports to `.ics`. The course forbids supervised methods, so we ask: *can RL with a verifiable reward alone teach a 0.5B model this?*

II. RL Formulation

State. $s_t = (x_{\text{sys}}, x_{\text{email}}, d_0, y_{<t})$: system prompt, email ($\leq$768 prompt tokens), received date d_0 and the tokens generated so far, embedded by the policy as $n \times 896$ hidden states; fully observed. **Action.** Discrete: the next token $a_t \in \mathcal{V}$, $|\mathcal{V}| = 151{,}936$. **Transition.** Deterministic append, $s_{t+1} = s_t \circ a_t$; the episode ends at <|im_end|> or after 320 tokens; γ=1; the reward is sparse and terminal. **Policy**: Qwen2.5-0.5B-Instruct [1] with LoRA [2] (r=16); the frozen base model is π_{ref}.

A. Reward

Let P, G be the predicted and gold events. If the output is not a schema-valid JSON list, $r_{\text{fmt}}=-1$ and nothing else pays; otherwise $+1$. For a pair (p, g) with $\delta = \mathbb{I}[p.\text{date}{=}g.\text{date}]$:

$$S(p,g) = 0.35\,c(p,g) + \big(\delta + 0.25(1-\delta)\big) \cdot \big(0.20\,s_{\text{time}} + 0.25\,F_1^{\text{title}} + 0.10\,s_{\text{type}} + 0.10\,s_{\text{loc}}\big) \quad (1)$$

with the date verifier $c = 1$ if δ=1, else $0.5\max(0, 1 - |\Delta d|/14)$ (Δd = error in days, computed by calendar arithmetic): "this Friday" vs. "next Friday" is told apart from a random date. Hungarian matching [3] gives $r_{\text{match}} = \sum_M S / \max(|P|, |G|)$, so misses and hallucinations both cost reward. $r_{\text{empty}} = +1$ for [] on a no-event email, -1 for hallucinating one, -0.5 for [] when events exist; $r_{\text{len}}=-1$ for any text around the JSON. The total is

$$R = 0.5\,r_{\text{fmt}} + 2\,r_{\text{match}} + r_{\text{empty}} + 0.25\,r_{\text{len}} \in [-1.75,\ 2.5]. \quad (2)$$

Incentives. The format gate makes parseable output the first thing learned; the date is the largest share of r_{match} and gates the other fields; normalising by $\max(|P|, |G|)$ makes spamming and skipping equally costly; a fake calendar entry costs more than a missed one. **Reward hacking fixed:** originally the title "a" scored 2.45 of 2.5 and a wrong-day event 1.8; token-F1 titles and date gating lower these to 2.0 and 0.83 (unit-tested); a wrong date still beats [] (= 0).

III. Environment and Data

Our own generator produces emails with exact labels at five levels used as a curriculum: L1 explicit dates; L2 varied formats; L3 relative dates; L4 postponements, reply chains and multi-event digests; L5 forwarded, noisy, Hinglish. A quarter of the templates appear *only* in the 300-email test split (train 2000, val 200), so the test measures generalisation; 40 test emails contain no event. An episode samples an email from the curriculum stage (first 100 steps L1–L2, then all levels), with 30–40% no-event emails in the final configurations. **All results are on synthetic data**: we did not label a real test set.

IV. Algorithm

Q-learning/DQN need a value per token action ($1.5{\times}10^5$ of them) and PPO a critic as large as the policy; GRPO [4] uses a group baseline instead: for each email we sample G completions and use $\hat{A}_i = (R_i - \text{mean}_j\, R_j)/(\text{std}_j\, R_j + \epsilon)$ for all tokens of completion i, maximising (TRL [5], token-normalised)

$$\mathcal{J} = \mathbb{E}\Big[\frac{\sum_{i,t} \min(\rho_{i,t}\hat{A}_i, \text{clip}(\rho_{i,t}, 1{\pm}0.2)\hat{A}_i)}{\sum_i |y_i|} - \beta\,\mathbb{D}_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}})\Big]. \quad (3)$$

The final configurations use G=8, 4 emails per step, temperature 1, learning rate $2{\times}10^{-5}$ (cosine, 50 warm-up steps) and 500 steps (≈3 h on one 6 GB GPU; the reference policy is the base model with the adapter off).

V. Experiments

Baselines: a regex+`dateparser` rule system, and zero-/3-shot prompting of the same 0.5B model and of Qwen2.5-1.5B with both prompts (constant [] and random guesses score 0.0

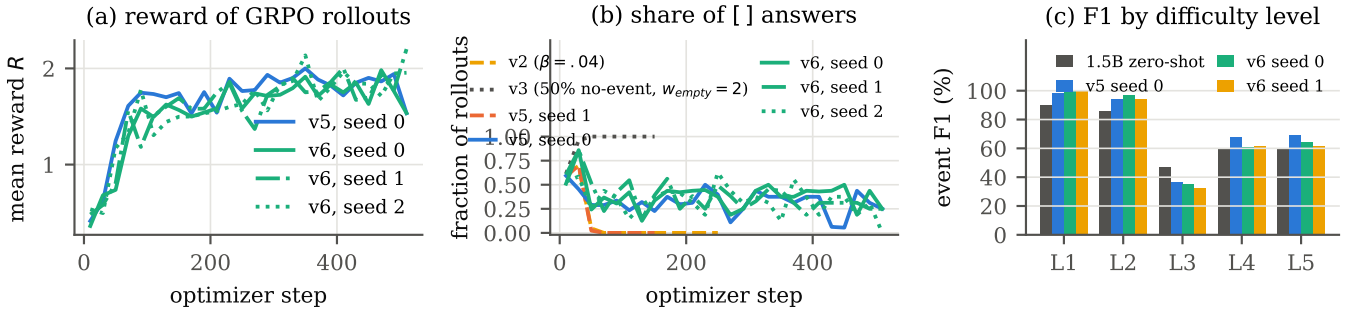


Fig. 1. (a) Mean rollout reward. (b) Share of answers equal to []: v2 and v5 seed 1 lose it within 50 steps, v3 answers only [], v5 seed 0 and v6 keep it. (c) Event F1 per level.

TABLE I

SYNTHETIC TEST SET (300 EMAILS, 40 WITHOUT EVENTS), %. PROMPT: 1 = ORIGINAL SYSTEM PROMPT, 2 = COMPACT JSON PROMPT WITH WORKED EXAMPLES (ALSO GIVEN TO THE BASELINES). NO-EVENT: SHARE OF THE 40 NO-EVENT EMAILS ANSWERED WITH []. GRPO ROWS ARE SINGLE RUNS; THE CONFIGURATION LETTERS ARE DEFINED IN SEC. V.

Method	Prompt	F1 [95% CI]	Date	JSON	No-event	L3 F1
Regex + dateparser	–	60.5 [54.2, 66.5]	62.7	100.0	67.5	68.2
Qwen-0.5B zero-shot	1	51.5 [46.2, 57.3]	54.6	84.0	0.0	19.6
Qwen-0.5B zero-shot	2	24.0 [18.2, 29.8]	23.5	90.3	100.0	9.5
Qwen-0.5B 3-shot	1	32.8 [28.7, 37.1]	43.8	74.0	22.5	11.1
Qwen-1.5B zero-shot	1	59.2 [53.6, 64.9]	70.9	97.3	7.5	42.5
Qwen-1.5B zero-shot	2	67.1 [61.8, 72.4]	82.4	95.3	15.0	46.8
A: baseline reward design, β=.04	1	61.4 [56.1, 67.1]	64.7	95.7	0.0	31.9
A + T=1, G=8	1	63.7 [58.4, 68.8]	68.6	96.3	0.0	31.9
A without curriculum	1	58.9 [53.5, 64.4]	67.0	94.7	0.0	32.1
B: new prompt, β=.3	2	65.8 [60.3, 71.6]	75.5	99.7	100.0	34.0
C: B with β=.15 (seed 0)	2	72.2 [67.4, 77.2]	79.4	99.3	100.0	36.2
D: C + 40% no-event (seed 0)	2	69.9 [64.6, 75.1]	77.5	98.0	100.0	35.3
D: C + 40% no-event (seed 1)	2	68.4 [63.3, 73.6]	78.1	99.3	100.0	32.1
D: C + 40% no-event (seed 2)	2	70.9 [65.9, 76.0]	79.4	100.0	100.0	34.3

F1). No SFT baseline: it is forbidden. An event counts as correct when its date is right and its title token-F1 is $\geq$0.5; intervals are 95% bootstrap CIs.

Results (Table I). The best run (C, seed 0) reaches 72.2% F1, 79.4% date accuracy and 99.3% valid JSON. The same 0.5B model scores 24.0% zero-shot (51.5% with the original prompt). The best run also exceeds the zero-shot 1.5B model (67.1%), but the intervals overlap, so we do not claim a significant win. D, the one configuration that kept [] on every seed, reaches 69.7$\pm$1.0% over 3seeds. Runs A (original prompt) reach only 61.4–63.7% and answer [] on 0 of 40 no-event emails: the base model never emits [], so GRPO never samples it and the policy hallucinates an event for every no-event email. The regex baseline (60.5%) is strong here and beats the policy on L3 relative dates.

The [] collapse (Fig. 1b). The compact prompt with [] examples and oversampled no-event emails fix exploration, but we found three attractors. *v2* (β=.04, 30% no-event): the policy first over-abstains, then within $\sim$50 steps stops producing [] on any email. *v3* (50% no-event, w_{empty}=2): the opposite, always [], because at those weights abstaining outearns extracting. A group of G answers that are all [] on a no-event prompt has zero advantage, so nothing protects the behaviour while updates from event emails leak into it. A stronger KL anchor (B, β=.3) fixes it (100.0% on no-event emails) but pins the policy too close to the base model (F1 65.8%). β=.15 (C) trains well on seed 0 but *collapsed on seed 1* at step $\sim$40, so the good result depends on the seed. D adds more no-event prompts (40%): [] survives on all 3seeds (Fig. 1b) and every no-event email is answered correctly, at an F1 of 69.9, 68.4and 70.9% for seeds 0, 1 and 2, all within the CI of C. We cannot tell whether the 40% mix or luck made the difference.

Failure analysis. Relative dates remain the weak spot in every run: on L3 emails the date is exact for 36.5% of events, off by one day for 30.8%, and off by four or more days for 26.9%. A typical error: "scheduled *tomorrow* at 19:00" (received 19 Aug) is answered 19 Aug, the received date. Partial credit does not teach calendar arithmetic in 500 steps.

Limitations. Synthetic emails only; C has one usable seed, and B–D were chosen after seeing test results, so numbers are optimistic.

Conclusion. Pure GRPO takes Qwen2.5-0.5B from 24.0% to 72.2% event F1 on unseen synthetic templates without any supervised data. The main lesson is that a rarely sampled behaviour (answering []) is silently lost or over-learned unless it is watched; the shipped adapter is run C.

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